4.0 Virtualization and Cloud Computing

4.1 Virtualization Concepts

Virtualization allows you to run multiple computing environments on a single physical system. This is done using software called **hypervisors**.

Purpose of Virtual Machines (VMs)

Purpose	Description & Example
Sandboxing	Isolated environment to safely test apps without affecting the host system.
Test & Development	Developers test software in different OS versions without needing extra hardware.
Application Virtualization	Run applications without installing them on the host OS.
Legacy Software/OS	Run old software on modern OS using a VM.
Cross-Platform Virtualization	Run Windows apps on a Mac or Linux using virtualization tools.

Requirements for Virtualization

1. **Security**
   - VMs must be isolated from the host to prevent malware spread.
2. **Network**
   - Virtual networks allow communication between VMs and external networks.
3. **Storage**

    - Requires fast, reliable storage for VM images and snapshots.

Desktop Virtualization

- **Virtual Desktop Infrastructure (VDI)**
    - Users access a desktop hosted on a central server.
    - Example: Employees log into their company desktop from home via VDI.

Containers

- Lightweight virtual environments.
- Run applications with their dependencies.
- Examples: Docker, Kubernetes.
- Difference from VMs: Share host OS kernel, faster startup, smaller footprint.

Hypervisors

Type	Description	Examples
Type 1 (Bare Metal)	Runs directly on hardware. No host OS required.	VMware ESXi, Microsoft Hyper-V, Xen
Type 2 (Hosted)	Runs on top of a host OS. Slower than Type 1.	VirtualBox, VMware Workstation

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4.2 Cloud Computing Concepts

Cloud computing provides on-demand computing resources over the internet.

Common Cloud Models

Model	Description	Example
Private Cloud	Infrastructure dedicated to one organization.	Company hosting its own cloud server.
Public Cloud	Infrastructure shared across multiple organizations.	Amazon Web Services (AWS), Google Cloud.
Hybrid Cloud	Combines private and public clouds.	Sensitive data in private cloud, web app in public cloud.
Community Cloud	Shared by organizations with similar needs.	Universities sharing research infrastructure.

Cloud Service Models

Service	Description	Example
IaaS (Infrastructure as a Service)	Provides virtualized hardware resources.	AWS EC2, Google Compute Engine
PaaS (Platform as a Service)	Provides platforms for development without managing hardware.	Google App Engine, Heroku
SaaS (Software as a Service)	Provides fully managed applications.	Microsoft 365, Gmail, Salesforce

Cloud Characteristics

Characteristic	Description & Example
Shared vs. Dedicated Resources	Cloud can share CPU, memory, and storage among users or dedicate them.

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Characteristic	Description & Example
Metered Utilization	Pay for what you use; includes **ingress/egress** (data in/out).
Elasticity	Scale resources up or down as needed. Example: Auto-scaling web servers.
Availability	Cloud services are highly available with redundancy and failover.
File Synchronization	Cloud keeps files synced across devices (e.g., OneDrive, Google Drive).
Multitenancy	Multiple users share infrastructure securely without accessing each other's data.

Key Comparisons

Aspect	Virtualization	Cloud Computing
Hardware dependency	Yes, runs on physical hardware	Yes, but abstracted; provider manages hardware
Resource sharing	Within one physical host	Across multiple hosts/data centers
Access	Local network or device	Internet-based
Purpose	Testing, legacy apps, isolation	Scalable services, remote access, shared applications

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Examples in Practice

- **Virtualization:** Running Ubuntu Linux in VirtualBox on a Windows laptop.
- **Cloud:** Hosting a web application on AWS with auto-scaling and global availability.

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